

Extra Credit Homework

- Submit a single PDF containing scans of your complete **hand-written** solutions and analysis, with all steps clearly shown.
- All calculations must be done by hand.
- Please upload a single, high-quality PDF file containing clear pictures of your handwritten work.

Exercise 1

- Perform two steps of gradient descent starting from the origin ($\mathbf{b}^{(0)} = (0, 0), c^{(0)} = 0$) for a linear regression problem in $\mathbb{R}^2$ with the following data points:

$$\mathbf{x}_1 = (1, 1), \quad y_1 = 1, \quad \mathbf{x}_2 = (1, 2), \quad y_2 = 2, \quad \mathbf{x}_3 = (2, 1), \quad y_3 = 2.$$

The goal is to minimize the mean squared error (MSE) for the linear model:

$$f(\mathbf{x}) = \mathbf{b}^T \mathbf{x} + c = b_1 x_1 + b_2 x_2 + c,$$

where $\mathbf{b} = (b_1, b_2)$ is the parameter vector, and c is the bias. Suppose the learning rate is 0.1 in all the iterations!

- Perform two steps of mini-batch gradient descent with a batch size of 2:
 - First batch: Points $(\mathbf{x}_1, \mathbf{x}_3)$.
- Perform two steps of gradient descent using a single point at each iteration:
 - First iteration: Use the third point $\mathbf{x}_3$,
 - Second iteration: Use the second point $\mathbf{x}_2$.

For each scenario, compute and update the parameters $\mathbf{b}$ and c over the two iterations.

Exercise 2

You are given the following linear programming problem:

Objective Function:

Maximize

$$-x_1 + 2x_2$$

Constraints:

$$x_1 + x_2 \leq 1, \quad x_1 \geq 0, \quad x_2 \geq 0$$

Solve this linear program using Karush-Kuhn-Tucker (KKT) conditions. You do not need to find the dual problem!

Exercise 3

You are given the following linear programming problem:

Objective Function:

Maximize $x_1 + 4x_2$

Constraints:

$$x_1 + x_2 \leq 2, \quad 2x_1 + x_2 \leq 4$$

Find the dual of this linear programming problem. You do not need to solve the problem itself or its dual!

Exercise 4

You are given the following quadratic programming problem:

Objective Function:

Minimize $\frac{1}{2} (4x_1^2 + x_2^2) + x_1 + x_2$

Constraints:

$$x_1 + x_2 \leq 2, \quad x_2 \geq 1$$

Solve this quadratic program using Karush-Kuhn-Tucker (KKT) conditions. You do not need to find the dual problem!

Exercise 5

You are given the following quadratic programming problem:

Objective Function:

Minimize $\frac{1}{2} (x_1^2 + 2x_2^2) - x_1 - 2x_2$

Constraints:

$$x_1 + x_2 = 4, \quad x_1 \geq 0$$

Find the dual problem. You do not need to solve the problem itself or its dual!

Exercise 6

You are given the following symmetric matrix:

$$A = \begin{pmatrix} 5 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

Compute the eigenvalues explicitly to determine the positive definiteness of this matrix!